

SUPERMASSIVE BLACK HOLES IN THE CENTER OF GALAXIES

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Abstract

Supermassive black holes reside in most local galaxies and play a key role in structuring them. This article presents the definition of a supermassive black hole and general information to understand its functioning in the center of galaxies, as well as its structures, hypothesis of formation, specific characteristics, and questions that are still on the agenda to be solved by the scientific community. From the confirmation of the existence of black holes predicted by Albert Einstein to the future of this spatial region, fundamental points are addressed to understand how BNSMs (Supermassive Black Holes) interact with the matter around them, including with light, which, according to Newtonian mechanics, does not have the necessary escape velocity to get out of the black hole.

In 1915, when the German physicist Albert Einstein elaborated the Theory of General Relativity - which shows that, for inertial references, space and time are relative and depend on the movement of the observer - a new possibility appeared in the world of science that would give towards research and questions, such as the existence of black holes. This subject was the focus of many debates and discoveries about its veracity until the moment when the first black hole was detected. Scientists know of its existence because the gravitational pull of the black hole is extreme, fragmenting the star into thin streams of material that get very close together and can send vibrations through gravitational waves that propagate through space-time out of the black hole at the speed of light. (HAWKING, 2017)

The direct detection of these vibrations is recent. On September 14, 2015, a gravitational wave signal with a frequency less than 10 Hz called GW 150914 was inferred. These gravitational waves originated from the merger of two black holes more than 1 billion years ago and traveled through space until they arrived on Earth at a size a thousand times smaller than that of a proton, approximately 0.8768×10^{-18} meters. In view of the succinct number of studies that address this topic to reach the general public, this article intends to carry out a bibliographical review of the major discoveries regarding supermassive black holes.

Keywords: supermassive black holes, galaxies, general relativity, gravitational waves, space-time.